

Game Development Experience

Directed Focus Study: *Street Fighter 3: 3rd Strike Clone*

Role: Sole Developer

- Created input buffers, state machines, and comprehensive animation system in a **Custom C++ engine**.
- Wrote a system that calculates custom timescale for different phases of an animation (such as startup, recovery, etc.) to ensure that those phases will play out in a developer-defined timeframe, measured in frames, while maintaining a smooth animation throughout.
- Implemented internal developer tools such as frame-by-frame time-stepping and dynamic **hurtbox/hitbox and debug visualization** for ease of use when tuning.
- Implemented support for local multiplayer on keyboard, controller, and **arcade stick**.
- Adhered to a strict schedule to ensure deliverables were ready by their corresponding deadlines.

Hamster Ballin'

Role: Race Logic Programmer

- Worked as part of a 43-person team to create a kart racing game.
- **Co-wrote** a system that maintained data persistence between levels in Unreal Engine 5 for a Grand Prix game mode. This system tracked overall score throughout the Grand Prix, for multiple local players and game AIs by initializing only the number of AI needed to fill a race, while also awarding points to the correct player or AI depending on how they placed at the end of a race in the Grand Prix.
- Set up a system for handling **kart-racer style pickups** that was later handed off to and finished by another team member.
- Created a **checkpoint and respawn system** for determining validity of a player's completed race lap and preventing unintended skips across a track.

Net Chess 3D

Role: Sole Developer

- Wrote a networking system that uses TCP/IP to perform LAN communication between devices to play Chess online with another device.
- Implemented a Blinn-Phong lighting shader with techniques for sampling normal maps, specularity, glossiness, and emissive.

- Wrote a custom .obj loader and implemented a third party .glTF loader (fastglTF) for loading and parsing 3D model files and generating meshes, surface/vertex normals, and surface/vertex tangents and bitangents.
- Wrote a system for determining the legality of proposed moves in Chess. This system was used to create a Chess game that could be played across a local network.

Doomenstein

Role: Sole Developer

- Wrote a system for measuring beats in a song via a provided bpm and polling the FMOD DSP clock to get playback time in samples. This system was used to create a beat shooter system where shots were judged and scored based on how close they were to a beat in the song.
- Created a system for spawning, managing, and deleting actors via bitflags and handles.
- Wrote a function for raycasting against a 3D grid map and custom cylinder colliders that returned the closest result to the ray origin. This was used to create hit-scan weapons.

Gravebound

Role: UI Programmer

- Worked as part of a 37-person team to create a roguelike first person shooter.
- Implemented a menu system in Unreal Engine 5 for handling **pause, settings, and displaying in-game information** (inventory, upgrades, etc.)
- Wrote a menu system that created shape-shifting buttons for handling **randomly determined upgrades**. Buttons determined their identities on creation and then destroyed themselves once an upgrade was chosen. When a button was clicked it would tell another team member's system about itself and to upgrade the player.

Education

Southern Methodist University Guildhall

August 2025 - May 2027

- Master of Interactive Technology
- 3.83 Cumulative GPA

University of Texas at Dallas

August 2020 - May 2025

- Bachelor of Arts - Arts, Technology, and Emerging Communication
- 3.69 Cumulative GPA

Work Experience

ID Tech - On-Campus Instructor

May 2025 - August 2025

- Taught students how to program and design games in Unreal Engine 5, GDevelop, and Roblox.
- Assisted in pickup and drop-off processes for students.
- Worked with students to create a deliverable product in a 1-week timespan